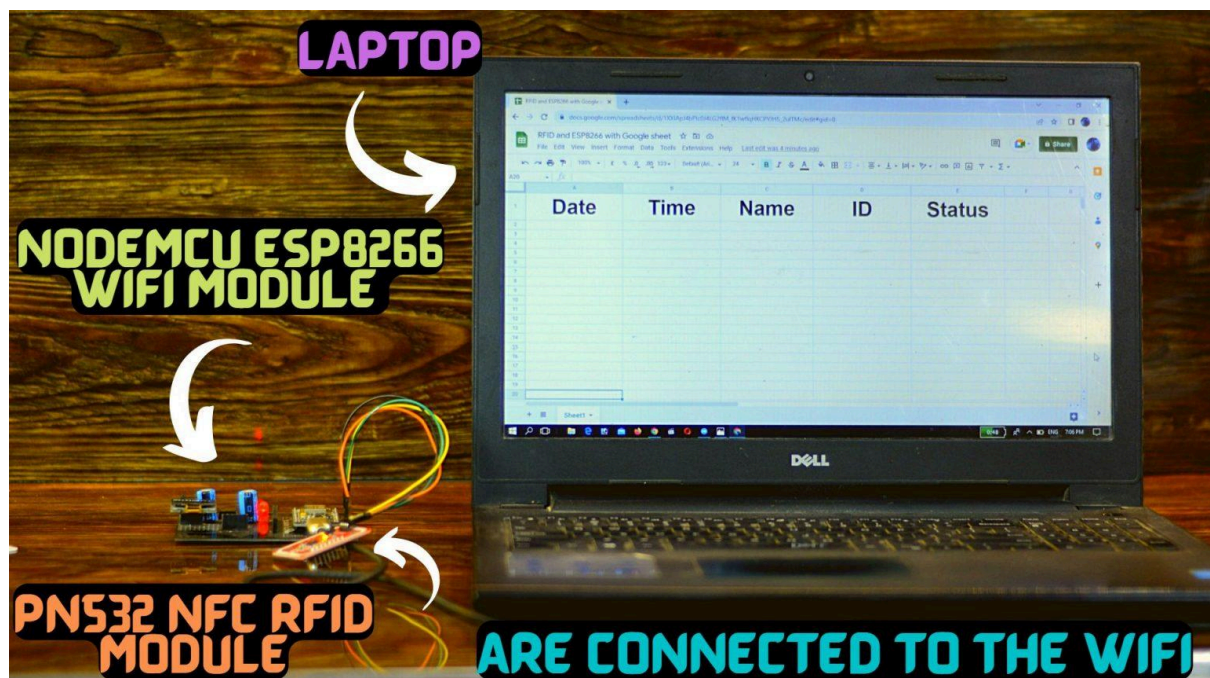


This project utilises RFID/NFC technology to create a real-time attendance system that updates data directly into Google Sheets. This allows users to view and track their attendance effortlessly, making it an ideal solution for educational institutions, businesses, and events. The system is cost-effective, as NFC/RFID tags are cheaper than other attendance systems, and it provides a seamless way to manage attendance data.



2. Competitive Analysis

Feature	Traditional Attendance Systems	Biometric Fingerprint Attendance Systems	NFC/RFID Attendance System with Google Sheets
Referral Pic			
Cost	Medium to High	High	Low
Real-time cloud integration	No	No	Yes
Ease of use for tracking	Hard to maintain	Requires hardware and software setup	Simple Google Sheets interface
Maintenance and Scalability	Requires regular maintenance	Expensive and complex	Easily scalable and low-maintenance

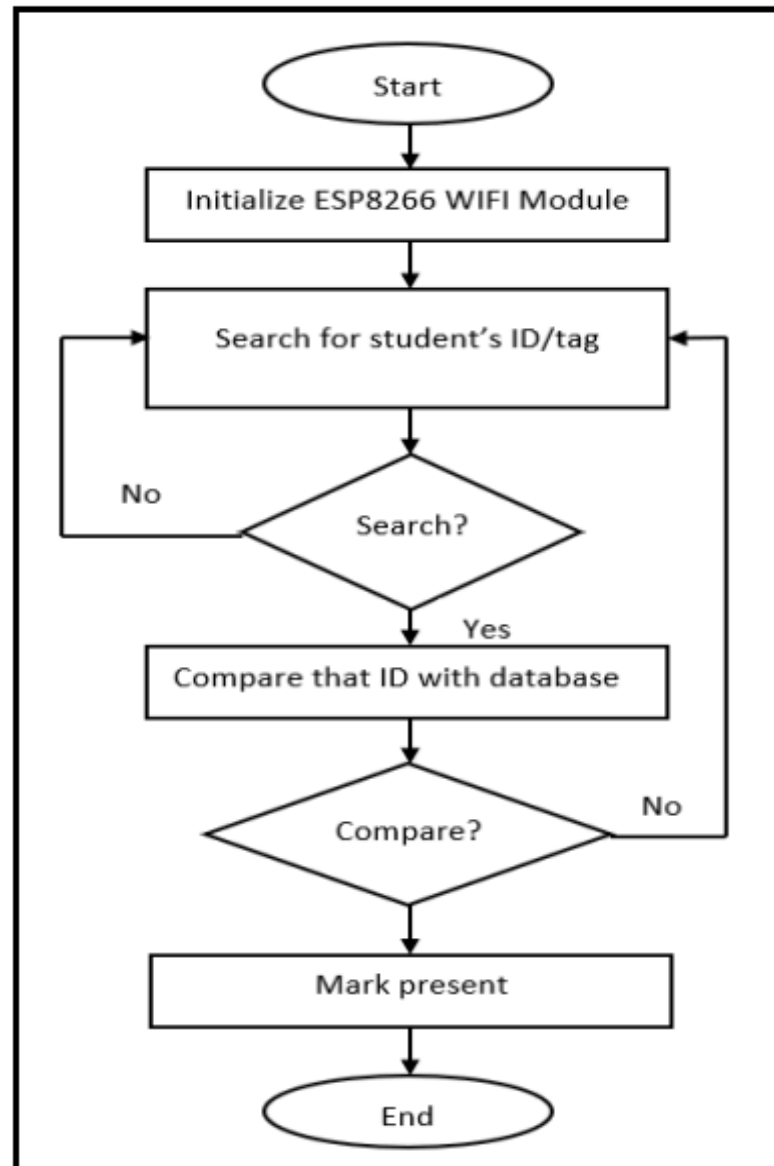
3. GAP Analysis

- High-Cost Attendance Systems:** Existing systems like biometric and RFID solutions are expensive due to the hardware and proprietary software involved.
 - Lack of Real-Time Cloud Integration:** Most systems do not offer real-time updates to cloud platforms like Google Sheets, making data management more complex.
 - Limited Accessibility:** Traditional systems often require special software or devices to view attendance data, which restricts accessibility.
 - Market Opportunity:** There is a gap for a low-cost, easily accessible system that offers real-time cloud data updates without requiring expensive software or hardware.
-

4. Flowchart

[Flowchart for NFC/RFID Attendance System]

1. User scans NFC/RFID tag
2. Tag ID is verified
3. Attendance is logged
4. Data is updated in Google Sheets in real-time
5. System ready for next scan

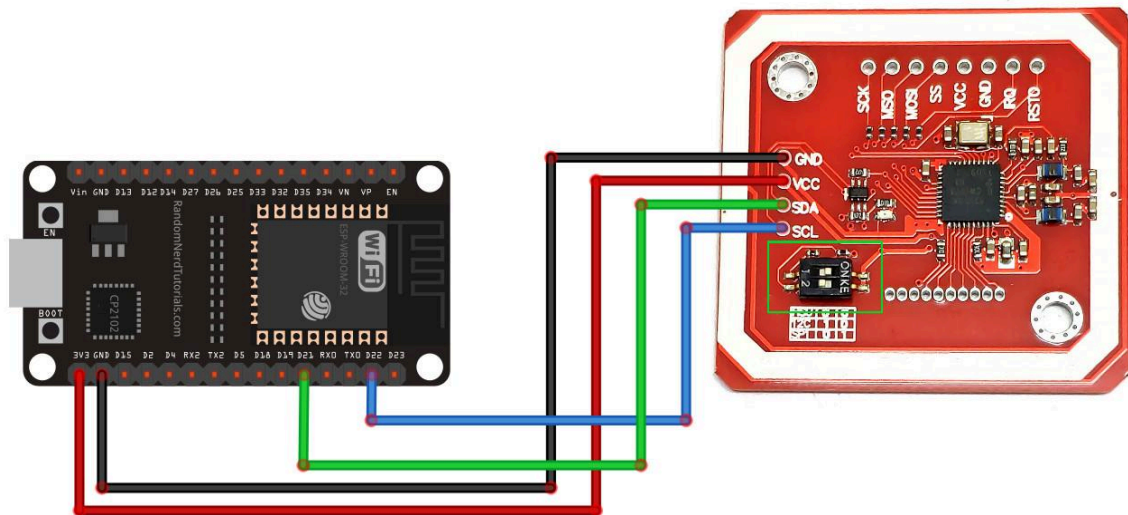


5. Problem Statement

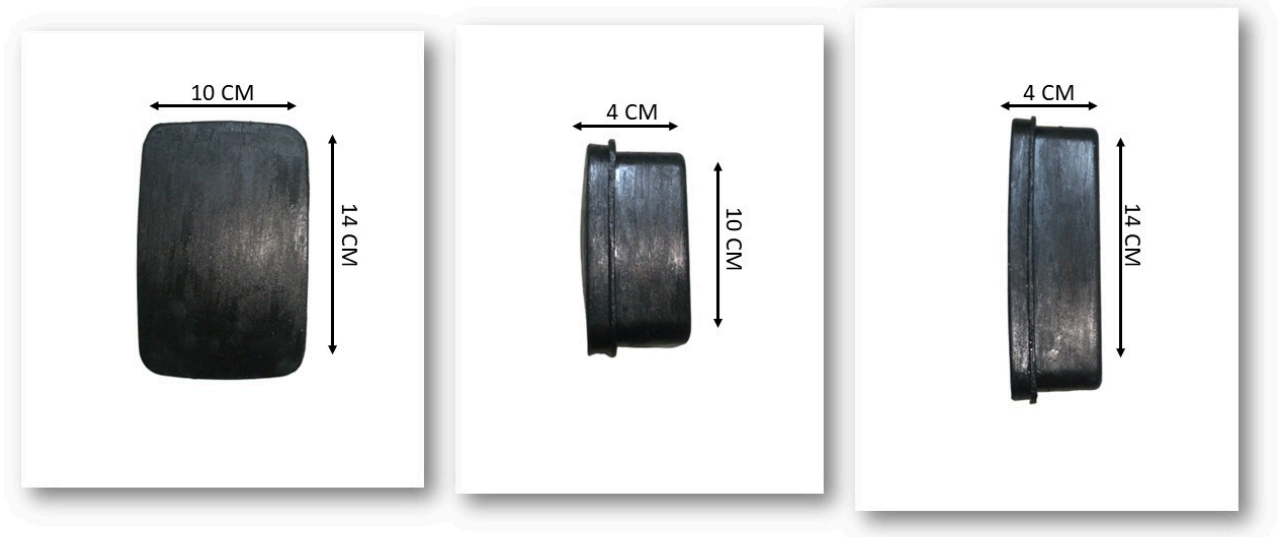
Ram needs a cost-effective and simple attendance system for his school that provides easy access to attendance data. He cannot afford a separate, complex database management

system, and thus, a solution that integrates Google Sheets and low-cost RFID/NFC tags is ideal for him.

6. Ideation / Circuit Diagram



7. Prototype:



7. List of Components (Specifications)

NodeMCU ESP8266

- **Purpose:** Wi-Fi enabled microcontroller for connecting to the internet and sending data to Google Sheets.
- **Specification:**
 - Voltage: 3.3V
 - Wi-Fi: 802.11 b/g/n
 - I/O Pins: 9 digital pins, 1 analog pin
 - Operating Frequency: 80MHz

PN532 NFC Module

- **Purpose:** NFC/RFID reader for scanning tags.
- **Specification:**
 - Interface: I2C
 - Operating Voltage: 3.3V - 5V
 - Range: 0-5 cm
 - Frequency: 13.56 MHz

Google Spreadsheet Integration

- **Purpose:** Real-time data storage and tracking of attendance.
- **Specification:**
 - Cloud-based storage system
 - Accessible from any device with internet access
 - Real-time data update through Google Scripts

Buzzer (3V - 6V)

- **Purpose:** Provides audible confirmation for successful tag scans.
- **Specification:**
 - Operating Voltage: 3V - 6V
 - Current: <20mA

8. Technical Features

1. **Wi-Fi Connectivity:** Uses NodeMCU for seamless connection to the internet.
 2. **Real-Time Data Logging:** Attendance data is updated instantly in Google Sheets.
 3. **NFC/RFID Technology:** Allows contactless attendance marking for improved user experience.
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9. Project Features

- 1. **Cost-Effective:** Uses inexpensive NFC/RFID tags and open-source tools.
 - 2. **Real-Time Access:** Attendance data can be viewed from anywhere via Google Sheets.
 - 3. **Scalability:** Can be easily scaled for larger institutions by adding more NFC readers.
-

5. Unit Testing

Feature	Before Unit Testing	After Unit Testing
Code Quality	High risk of bugs and errors in the code	Reduced bugs, high code reliability
Time for Debugging	Extensive time spent identifying and fixing bugs	Faster debugging process, minimal time spent
Code Coverage	Limited code coverage, potential untested scenarios	Comprehensive code coverage ensuring all functions are tested
Error Handling	Poor error handling with frequent unexpected crashes	Improved error handling with well-tested edge cases

10. Ways Forward

In the future, we plan to enhance the system by adding a **message alert system** that will notify users of their attendance status via SMS or email, providing an additional layer of convenience and accessibility.